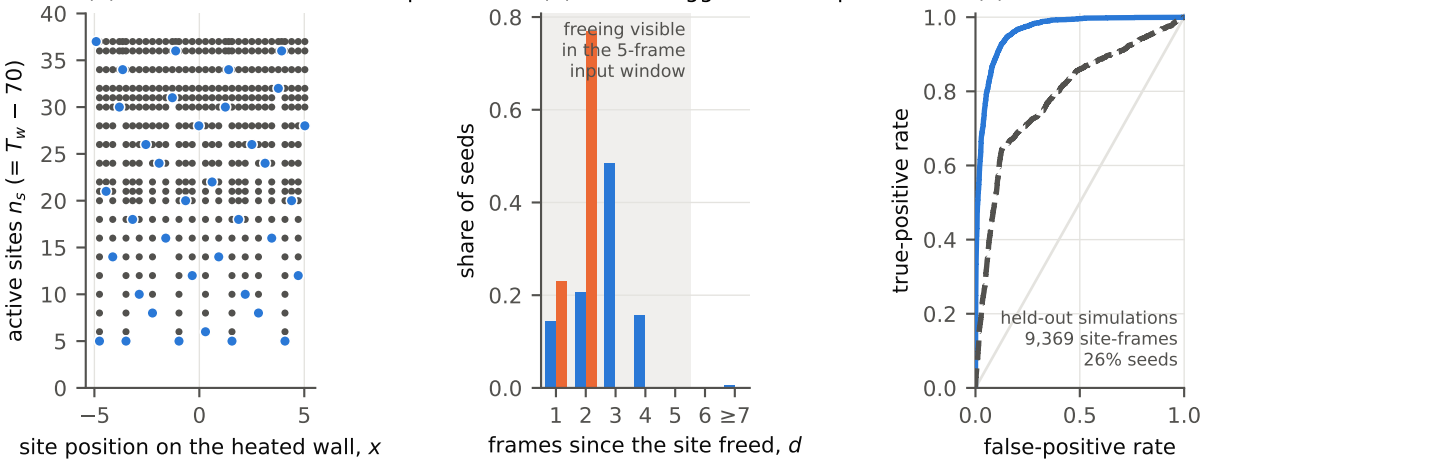
(a) Where: one fixed site sequence
active sites n_s (= T_w − 70)
site position on the heated wall, x
sites added at this T_w
sites already active below it
(b) When: trigger in the input
share of seeds
freeing visible in the 5-frame input window
frames since the site freed, d
pool boiling, 12–37 sites (99% inside)
single-bubble control (100% inside)
(c) Which frame: readable
true-positive rate
false-positive rate
held-out simulations
9,369 site-frames
26% seeds
local CNN on the input window: AUC 0.96
(within d = 3 / 4: 0.93 / 0.91)
delay since the site freed: AUC 0.79
(within d = 3 / 4: 0.48 / 0.50)